Audio-Visual Speech Recognition

Multimodal Interaction Project

Master's Degree in Computer Science

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SAPIENZA
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Why Audio-Visual Speech Recognition?

- Automatic Speech Recognition (ASR) degrades under real-world noise
- The **cocktail party effect**: overlapping voices mask speech formants
- **Key insight (McGurk effect):** the visual modality — lip movements — is *noise-invariant*
- AVSR combines audio and video to build a robust, complementary system

Goal

Classify 10 spoken Italian commands under 6 degradation conditions using an Audio-Visual fusion architecture.

Project Contributions

1. **Multimodal Early Fusion architecture:**
   Audio-LSTM + Video-BiLSTM branches fused at feature level
2. **Systematic elimination of experimental biases:**
   Padding Leakage (Continuous Noise Canvas) and Data Leakage (Stratified Group K-Fold)
3. **6-condition Ablation Matrix:**
   Empirical validation of modality orthogonality, multimodal synergy, and negative transfer

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Dataset

- **10 Italian spoken commands:**
  avvia, *stop*, *sopra*, *sotto*,
  sinistra, *destra*, *apri*, *chiudi*, *si*, *no*
- Synchronized **WAV** (audio) + **MP4** (video) recordings
- Single speaker, controlled environment
- Augmentation pipeline:
  - Babble Noise at +5 dB and −15 dB SNR
  - Spatial Jitter (high-frequency perturbation)
  - Static Tilt (off-axis framing)

Augmentation Conditions

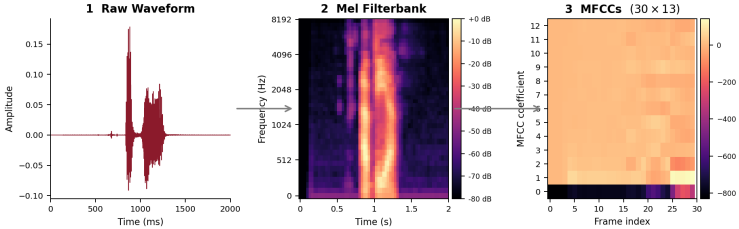
Modality	Type
Audio Light	+5 dB Babble
Audio Heavy	−15 dB Babble
Video Light	Jitter
Video Heavy	Tilt
A+V Light	Both

Audio Feature Extraction

2 Data & Features

- Raw audio resampled to **16 kHz**
- **13 MFCCs** computed per frame
- Fixed sequence length: **30 frames**
- Input tensor shape: $(30, 13)$

- **Why MFCCs?**
  - DCT decorrelates spectral envelope
  - Speaker-independent representation
  - Isolates phonetic formants from raw energy

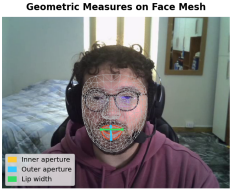


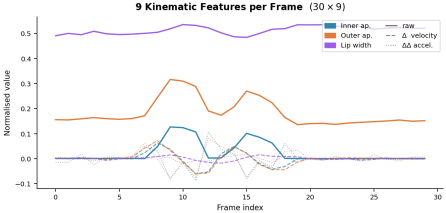
Video Feature Extraction

2 Data & Features

- **MediaPipe Face Mesh** detects 3D facial landmarks
- 3 static geometric measures per frame: inner aperture, outer aperture, lip width
- **Problem:** static geometry $\Rightarrow$ gradient stagnation

- **Solution:** kinematic derivatives
  - Δ (velocity) and $\Delta\Delta$ (acceleration)
  - $3 \rightarrow 9$ features per frame
- Input tensor shape: $(30, 9)$





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Bias #1: Data Leakage

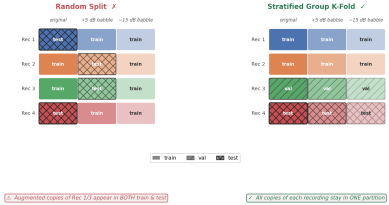
3 Methodology

Problem

Random split ⇒ augmented variants of the same recording appear in *both* train and test sets. Result: **100% accuracy** even at -15 dB — artificially inflated by speaker memorization.

Solution: Stratified Group K-Fold

1. `GroupKFold` (n=7): carve fixed **test** set
2. `GroupKFold` (n=5): **train / val** split
3. Group = base filename; stratification enforces class balance



Bias #2: Padding Leakage

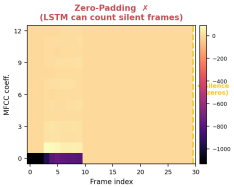
3 Methodology

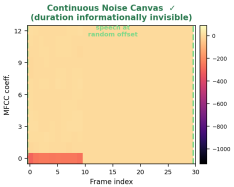
Problem

Zero-padding to 30 frames ⇒ LSTM *counts* zero-frames to infer spoken word duration. The network bypasses phonetic learning — it just counts silence.

Solution: Continuous Noise Canvas

1. Noise canvas of full target length (30 frames)
2. Speech waveform at a **randomized offset**
3. Every frame contains non-zero acoustic data





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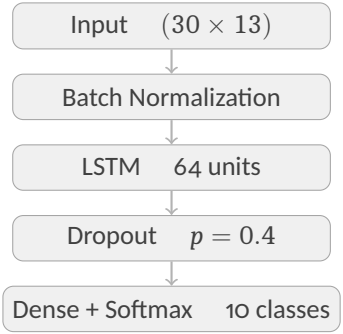
Audio-Only LSTM

4 Architectures

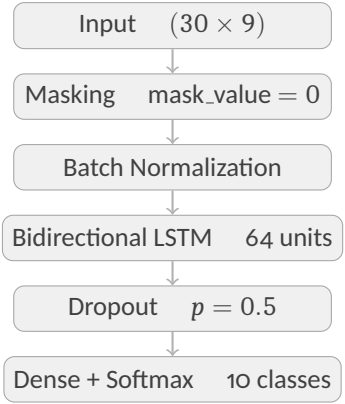
- Input: $(30, 13)$ MFCC sequence
- **Unidirectional LSTM:** speech formants unfold causally
- BatchNorm normalizes MFCCs online during training



Video-Only BiLSTM

4 Architectures

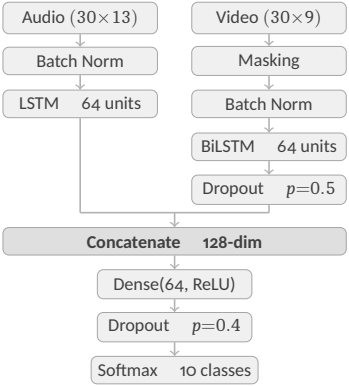
- Input: $(30, 9)$ kinematic sequence
- **Bidirectional LSTM:** lip movements exhibit coarticulation — future context is needed
- Masking ignores zero-padded frames



Early Fusion Multimodal Model

4 Architectures

Two parallel branches fused at feature level. The shared Dense layer learns **cross-modal correlations** (e.g. acoustic silence ↔ lip closure).



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Ablation Matrix

5 Results

Model	Clean	Aud. L	Aud. H	Vid. L	Vid. H	A+V L
Audio LSTM	100%	100%	44.83%	100%	100%	96.55%
Video BiLSTM	96.55%	96.55%	96.55%	41.38%	96.55%	34.48%
Early Fusion	**100%**	**100%**	**86.21%**	**96.55%**	**100%**	**96.55%**

- Aud. L/H = Babble Noise at +5 dB / -15 dB SNR
- Vid. L/H = Spatial Jitter / Static Tilt
- A+V L = simultaneous +5 dB noise + Jitter
- **Overall:** Fusion **96.55%** $>$ Audio 90.23% $>$ Video 77.01%

Multimodal Synergy & Dynamic Gating

Audio Recovery (Aud. Heavy)

- Audio LSTM: 44.83%
- Early Fusion: **86.21%**
- Recovery: $+\mathbf{41.38}$ pp

Visual features act as an anchor when audio is catastrophically degraded.

Video Recovery (Vid. Light)

- Video BiLSTM: 41.38%
- Early Fusion: **96.55%**
- Recovery: $+\mathbf{55.17}$ pp

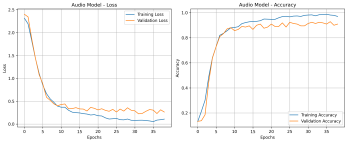
Network ignores corrupted kinematics and relies entirely on clean audio.

- **A+V Light:** fusion achieves 96.55% — no negative transfer, but no synergy when *both* channels are degraded simultaneously

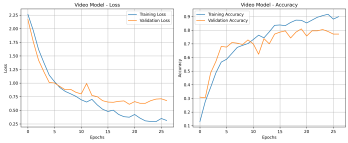
Learning Curves

5 Results

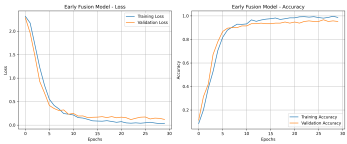
Audio LSTM



Video BiLSTM



Early Fusion

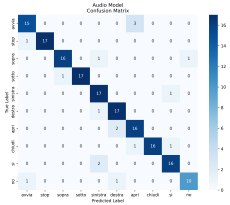


- **Audio:** stable convergence, val accuracy tracks training closely
- **Video:** slower and noisier convergence — geometric features carry less signal
- **Fusion:** fastest convergence — complementary modalities simplify the classification task

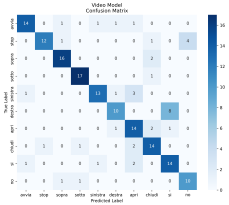
Confusion Matrices

5 Results

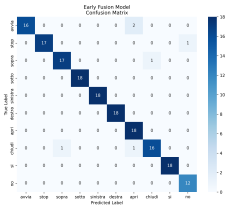
Audio LSTM



Video BiLSTM



Early Fusion



- Audio errors cluster in **Aud. Heavy** condition (-15 dB destroys formants)
- Video errors cluster in **Vid. Light** (Jitter destroys $\Delta/\Delta\Delta$ derivatives)
- Fusion has the **cleanest diagonal** — residual errors only in Aud. Heavy

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Conclusions & Future Work

6 Conclusions

Results

- Early Fusion: **96.55%** overall accuracy
- Audio-Only: 90.23% Video-Only: 77.01%
- Visual modality provides **+41 pp recovery** under -15 dB babble noise
- No negative transfer observed

Limitations

- Single speaker, controlled environment
- Simple concatenation (Early Fusion) cannot dynamically gate per-frame attention

Future Work

- **Cross-Modal Attention** and Transformer-based architectures for dynamic per-frame weighting
- Expand to multi-speaker, continuous speech vocabulary
- Edge deployment via model quantization

Audio-Visual Speech Recognition

Thank you for listening!
Any questions?